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CISC 5380 Programming with Python

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CISC 5380 Final Report: Manufacturer Attributes and Driver Fatalities

Introduction

When selecting the "correct" data for our report, we were particularly adamant on finding data that could help us gain insight on a practical issue opposed to complete hypotheticals as it would be intuitive when working through the modeling and analysis process. After completing individual queries, and synthesizing our results we decided to examine driver fatalities with respect to select vehicle manufacturers. The formal problem statement, as described in our proposal "Since 1966 the Department of Transportation and the National Highway Traffic Safety Administration have campaigned to maintain data reporting requirements relative to vehicular recalls. These reports include a consequence summary and the intended corrective action(s) on behalf of the car manufacturer. These efforts are in no small part a valuable contribution to the cumulative effort to increase traffic safety–and yet, fatal motor accidents have only been on a steady rise in the last decade." Our objective was to identify possible predictors for driver fatality rates utilizing the culmination of data analysis and data mining methodology acquired over the course of the semester. We hypothesized that variation in vehicle manufacturer will have a significant impact on driver fatalities; however also aim to possibly identify other contributing factors.

Data Preprocessing and Aggregation

Together Lily and I examined both the Department of Transportation Vehicle Recalls, and the National Highway Traffic Safety Administration Reports - Fatality Analysis Reporting System (FARS). Referring back to our potential questions from our proposal, we selected the various datasets that we would be able to work with. From the FARS reports, we decided on accident.csv, and vehicle.csv. In conjunction with the recalls_data.csv; Lily was able to complete the preprocessing stage of our assignment.

After checking for missing data (or lack thereof), there was no need to perform any missing data imputation. From here she was able to join our vehicle and recall datasets by way of a manufacturer key to ensure consistency in manufacturing and model names. We also had to account for the 3 year time lag in vehicle recalls, so she created a column with these lag accounted recall values which aggregates the recalls and fatalities by manufacturer which allowed us to create a Cars on the Road (COR) proxy for each manufacturer. Finally, she merged everything into a combined dataframe which is what we were able to utilize to perform the following analyses. For additional information on this stage, please refer to Lily's report.

Methodology

Pearson's Correlation Coefficient

First, we decided that we must not assume any degree of correlation and thought it would be best to run Pearson's Correlation (See Appendix A). Here, I am directly running 'lag account recalls' against 'deaths' with respect to manufacturers. Results (Appendix B) computed an r correlation of ~0.914, with a p-value of 0.00055. Simply, as we anticipated there is a strong positive correlation between recalls and fatalities. As recalls increase, the death trends rise along with it. Respectively our p-value is indicative of a statistically significant relationship, this

correlation is not by chance. However, correlation does not equal causation which is what permitted us to run data mining algorithms such as K-means and Linear Regression to see if we can find less obvious findings.

K-means Clustering

As a part of our overarching analysis, Lily independently performed K-means clustering where she delves into how manufacturer groupings provided insights into our project inquiry. She discusses our project narrative with respect to unique attributes about the manufacturers, having to pull from our limited domain knowledge of the automotive industry during interpretation. Refer to her report for actual result figures and analysis. For my independent report, I will only discuss her results in regards to my overarching interpretations and project outlooks.

Multiple Linear Regression (MLR)

I recommended that we run multiple linear regression opposed to simple linear regression as it could possibly reveal a more complex relationship within the data by creating predictions. By running multiple linear regression (see Appendix A) adding the predictor of COR, I was able to get some interesting results (see Appendix B). Starting with our recall coefficient at around 2.28×10^{-8} (0.0000000228) we can see that recalls do not hold significant weight in relation to fatalities. As recalls increase, the number of deaths do not increase to any meaningful extent. The COR coefficient tells a different story; at 8.26×10^{-5}, we can say roughly for every car on the road, our predicted fatalities will increase by about 0.0000826. Intuitively this makes sense; the more congestion there is on the word the more likely an accident is to occur per manufacturer – but this can be attributed to exposure rather than manufacturing error. Finally we have an r-squared score of 0.931, where 93.1% of the variation in fatalities between manufacturers can be

explained by recalls and cars on the road. However as we saw with our previous 2 coefficients, we can really attribute this to COR, meaning recalls add little to no predictive power in fatalities.

Simple Linear Regression (SLR)

Out of curiosity I decided to run a simple linear regression (see Appendix A) just to confirm my previous findings since it was easier to do using python packages. At first glance, these outputs (see Appendix B) seem meaningful but when compared to our findings from multiple regression, we know better. This model calculated a recall coefficient of 7.49×10^{-6} (0.00000749), lower than our multiple regression output; with a lower r-squared score of 0.836. We can support that recalls hold no predictive value to fatalities.

Model Comparison (MLR v. SLR)

We can even see the comparison in the predictive performance between the two models when looking at their "Actual v. Predicted Fatalities" (see Appendix B). We can see the distinct accuracy of their predictions; where SLR tends to overestimate fatalities, simply because its predictions are a function of recalls per manufacturer. MLR on the other hand does a better job of staying in line with the true value y (deaths), but admittedly does have a few outliers (e.g. Mercedes, Volkswagen).

Personal Interpretations

After synthesizing all our findings per model, I believe that there are some things to be noted about both our execution and our outputs. I think that overall individual model selections were up to par with our objective. Regression analysis provided the knowledge of how our features actually interacted with each other and how we can utilize one attribute to predict another. While K-means clustering actually showcases the structure of our aggregated datasets, showing that overall safety and fatalities are in some way tied to manufacturer type. Based upon

our dataset I can make the argument that it may have more to do with things like regionality, or customer demographics; at least more than we initially believed. It's not simply a function of actual manufacturing issues. Given that we found that COR are more predictive of fatalities than recalls are, I believe that location may be an eligible attribute to include. In our initial preprocessing, columns such as state and city were frequent but we unfortunately deemed them unimportant. I think If I were to rerun this project, I would attempt to quantify more geographic data; or even limit our data by state. Not surprisingly, our results weren't exactly what I was looking for, but they proved to be insightful enough to provide the foundation for another analysis as desired in any research cycle.

Limitations and Final Thoughts

Due to the complex nature of our dataset, time constraints, and limited experience with processing such large datasets; we had to significantly limit the number of features we were working with. As a result we also had to reduce our initial 3 questions within our proposal to the one stated in the introduction of this report. With that being said, I would recommend for future iterations to incorporate a broader range of features – possibility running these models are various combinations of features to compare their results. Due to our limited domain knowledge; there are likely various gaps and confounders in our analyses that we may not be aware of. Otherwise, I believe that our project produced meaningful results that we may be able to expand upon in the near future.

APPENDIX A

Code provided for each model mentioned, please note preprocessing and K-means clustering is not included here as that is Lily's independent work.

```
#PEARSON'S CORRELATION
recalls = df['Lag Accounted Recalls']
deaths = df['Deaths']
r,p = pearsonr(recalls, deaths)
print("Pearson correlation coefficient: ", r)
print("p-value: ", p)
```

Figure A1. Pearson's Correlation code using Pandas and Scipy.stats

```
#MULTIPLE LINEAR REGRESSION
X = df[["Lag Accounted Recalls", "Cars on Road"]]
y = df["Deaths"]

MLR_model = LinearRegression()
MLR_model.fit(X, y)

print("\nMLR Results (X = Recalls, Cars on the Road)")
print("y-intercept:", MLR_model.intercept_)
print("Coefficients:", MLR_model.coef_)
print("Recall coefficient:", MLR_model.coef_[0])
print("Cars on the Road coefficient:", MLR_model.coef_[1])

MLR_y_pred = MLR_model.predict(X)
print("R-squared score:", r2_score(y, MLR_y_pred))
```

Figure A2. Multiple Linear Regression code using Pandas, and Scikit-learn

```
# SIMPLE LINEAR REGRESSION (Recalls and Deaths only)
SLR_X = df[["Lag Accounted Recalls"]]
simple_model = LinearRegression()
simple_model.fit(SLR_X, y)

print("\nSLR Results (X = Recalls)")
print("y-intercept:", simple_model.intercept_)
print("Recall Coefficient:", simple_model.coef_[0])
SLR_y_pred = simple_model.predict(SLR_X)
print("R-squared Score:", r2_score(y, SLR_y_pred))
```

Figure A3. Simple Linear Regression code using Pandas, and Scikit-learn

```
# CALCULATING PREDICTIONS
df["Predicted_Deaths"] = MLR_y_pred
print("\nActual vs. Predicted Fatalities (MLR)") #header
print(df[["Manufacturer", "Deaths", "Predicted_Deaths"]])

df["Predicted_Deaths"] = SLR_y_pred
print("\nActual vs. Predicted Fatalities (SLR)") #header
print(df[["Manufacturer", "Deaths", "Predicted_Deaths"]])
```

Figure A4. Prediction Calculations for MLR v. SLR

APPENDIX B

Model outputs as produced by Pycharm, in order of appearance; K-means output not included, refer to Lily's independent report

```
Pearson correlation coefficient:  0.9145803157685164
p-value:  0.0005512650293930802
```

Figure B1. Pearson Correlation Results

```
MLR Results (X = Recalls, Cars on the Road)
y-intercept: -4.9186112071099615
Coefficients: [2.28266879e-08 8.26245232e-05]
Recall coefficient: 2.28266878742539e-08
Cars on the Road coefficient: 8.26245231682947e-05
R-squared score: 0.9311757419824976
```

Figure B2. Multiple Linear Regression Results

```
SLR Results (X = Recalls)
y-intercept: 100.81680987915615
Recall Coefficient: 7.492087907430339e-06
R-squared Score: 0.8364571539912395
```

Figure B3. Simple Linear Regression Results

Actual vs. Predicted Fatalities (MLR)

	Manufacturer	Deaths	Predicted_Deaths
0	BMW	278	294.922439
1	Chrysler	1590	1708.053817
2	Daimler	170	153.328051
3	Ford	1728	2012.093075
4	General Motors	2545	2490.716328
5	Honda	1490	1201.298165
6	Mercedes	166	311.842606
7	Nissan	1512	1100.389440
8	Volkswagen	345	551.356080

Figure B4. MLR Actual vs. Predicted Fatalities

Actual vs. Predicted Fatalities (SLR)

	Manufacturer	Deaths	Predicted_Deaths
0	BMW	278	410.418015
1	Chrysler	1590	1970.104872
2	Daimler	170	184.790873
3	Ford	1728	1889.757616
4	General Motors	2545	2089.203694
5	Honda	1490	1662.675953
6	Mercedes	166	344.813720
7	Nissan	1512	797.299777
8	Volkswagen	345	474.935478

Figure B5. SLR Actual vs. Predicted Fatalities